

Neural Data-Driven Captioning of Time-Series Line Charts

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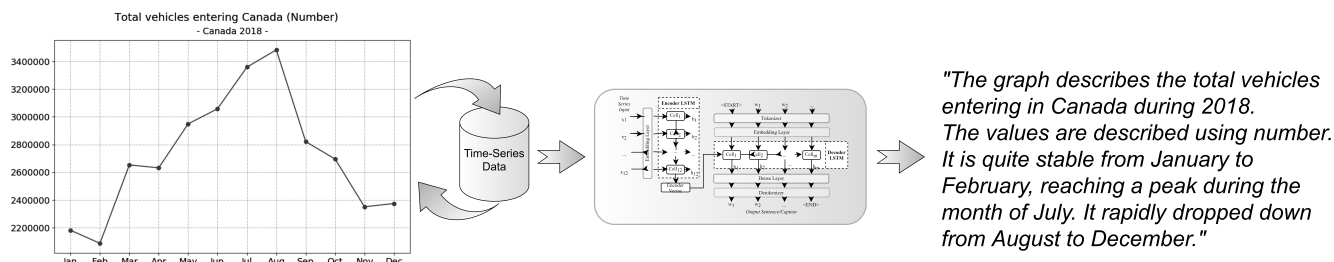


Figure 1: A sample caption generated by our model taking as input the same data used to create the shown line chart.

ABSTRACT

The success of neural methods for image captioning suggests that similar benefits can be reaped for generating captions for information visualizations. In this preliminary study, we focus on the very popular line charts. We propose a neural model which aims to generate text from the same data used to create a line chart. Due to the lack of suitable training corpora, we collected a dataset through crowdsourcing. Experiments indicate that our model outperforms relatively simple non-neural baselines.

CCS CONCEPTS

• **Computing methodologies** → *Natural language generation; Neural networks.*

KEYWORDS

Natural Language Generation, Machine Learning, Neural Network, Dataset, Visualization

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1 INTRODUCTION

Image captioning has greatly benefited from the neural revolution. Nowadays, very effective solutions based on the widely used

encoder-decoder architecture [2, 9] can be trained on large human labeled datasets [15, 22]. The ability to automatically generate captions for images can support several downstream tasks, including information retrieval and video summarization in systems for question answering and search engines. In this paper, we explore how similar text captioning solutions could be devised for the related and equally critical task of generating captions for information visualizations (infoVis). Although this problem has been tackled in the past by developing engineered captioning techniques [16], data-driven infoVis captioning has received limited attention. Only very recently [8] studied the generation of a caption starting from the bitmap of the visualization (e.g. to help visually impaired people), while [13] explored the related task of answering y/n questions about a given chart, again starting from the bitmap (e.g. to help extracting info from existing charts). In contrast in this paper, we investigate the generation of captions from the same data that was used to generate the chart, which would be useful in rather different scenarios, like for supporting data analysis by providing descriptions and insights about the data (e.g. complementing the approach in [23]) and/or in the generation of multimedia documents, as originally explored with engineered solutions in [10].

Since there are no existing datasets to train and test our data-driven captioning approach, our first contribution was to create a seed version of such dataset via crowdsourcing. Furthermore, as the space of possible visualizations is vast [18], it is unthinkable to deal with such complexity at once. For this reason, our first step towards infoVis captioning focuses on simple, but very popular visualizations, namely line charts, which usually represent the best way to show trends and changes over time. For instance, line charts are frequently used in finance in order to show variations in stock prices, in meteorology to show monthly rainfall and in many other fields to show similar time-series.

With respect to the proposed method, we frame the problem of infoVis captioning as data to text generation [4, 20], for which current neural solutions are trained in an end-to-end fashion using the very successful encoder-decoder architecture [3, 24]. In our

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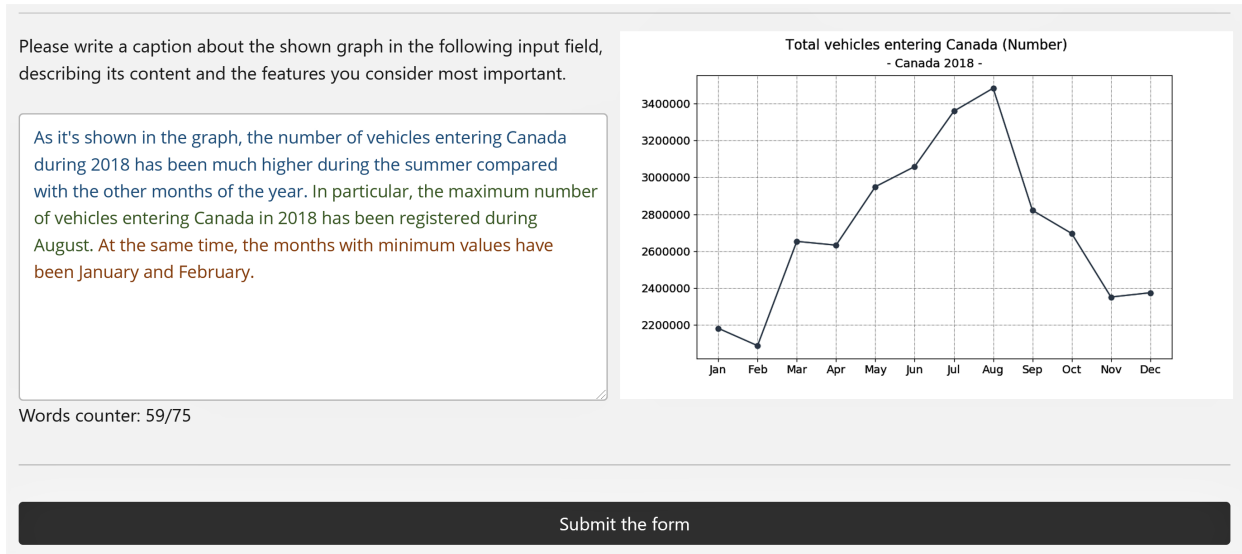


Figure 2: Annotation interface used to collect captions. A sample caption is shown on the left, with each single sentence in a different color. On the right, the line chart described by the caption.

proposal (see Figure 1 and 3), the input to the captioning system is the same that was used to generate the chart, including not just the time series values, but also information such as the title, the axis labels, geographic location, year and unit of measurement (UOM). This input is first normalized and generalized to better support the learning process. Then, it is fed into an encoder-decoder neural network model which gives as output a natural language description. We complete our study by running a series of experiments in which we compare our neural solution with non-neural baselines. At the end, we discuss the results of this evaluation and interesting insights we derived from their analysis.

2 NEW DATASET AND ANNOTATION UI

In order to study the new task of data-driven InfoVis captioning, a suitable dataset was needed to train and test possible alternative approaches. More specifically, such a dataset must include the following elements: (a) *Time series data*: composed by data values and general information as topic, year, geo location and unit of measurement. (b) *Data chart images*: line charts images generated using the collected time series data. (c) *Textual captions*: descriptions in Natural Language of the data chart image associated with the time series data.

Unfortunately, to the best of our knowledge, a dataset containing all the required elements does not exist. While it is easy to find datasets containing time-series which are publicly accessible on the web, for instance about weather data [19] and about stock time series[1], they never include natural descriptions (i.e., captions). Furthermore, they often lack variety. So we created a new custom dataset¹, which would arguably not just be useful to conduct the current work, but could also become a pilot dataset for future works and for a possible future wider data collection phase. The data gathering process consisted of three main steps.

¹https://github.com/Sprea22/NeuralCaptioningSystem_Model-Dataset

Step 1 - Time Series Data: Our data source for time series data is an online platform [19] which provides a significant quantity of public statistical data about different topics in Canada. We collected data about 100 different times series, including time series values, year, geographic location, frequency, units of measurement and title. We ensured a certain degree of heterogeneity between the collected time series calculating the Pearson correlation coefficient[5] between all of them and selecting a group of time series which are rather dissimilar from each other (Avg correlation of 0.041).

Step 2 - Line Chart Images: The collected time series data have been used to generate 100 line chart images. It was decided to display all the data as line charts (vs. bar charts) because this visual encoding is the initial focus of our current study. Furthermore, it is considered the most effective encoding for showing trends and changes over time [18].

Step 3 - Natural Language Captions: In order to collect captions for each line chart, we built a web interface² for crowdsourcing (see Figure 2). A large number of people was contacted, asking them to submit captions for our charts. In particular, each contributor was asked: "Please write a caption about the shown graph, describing its content and the features you consider most important". Each contributor, before proceeding with the captioning process, was presented with the following instructions: (a) Try to describe the chart using less than 75 words. (b) Accomplish the task between 3 and 10 times. (c) Click on the "Show vocabularies" button to see words and phrases that can be helpful in describing time series (e.g., fluctuate, plummet, boost). After that, the contributor was presented with a series of charts, one at the time randomly selected from the dataset, and asked to provide captions. English proficiency and level of education of the contributors were controlled in order to guarantee caption quality.

²https://github.com/Sprea22/NeuralCaptioningSystem_WebAppUI

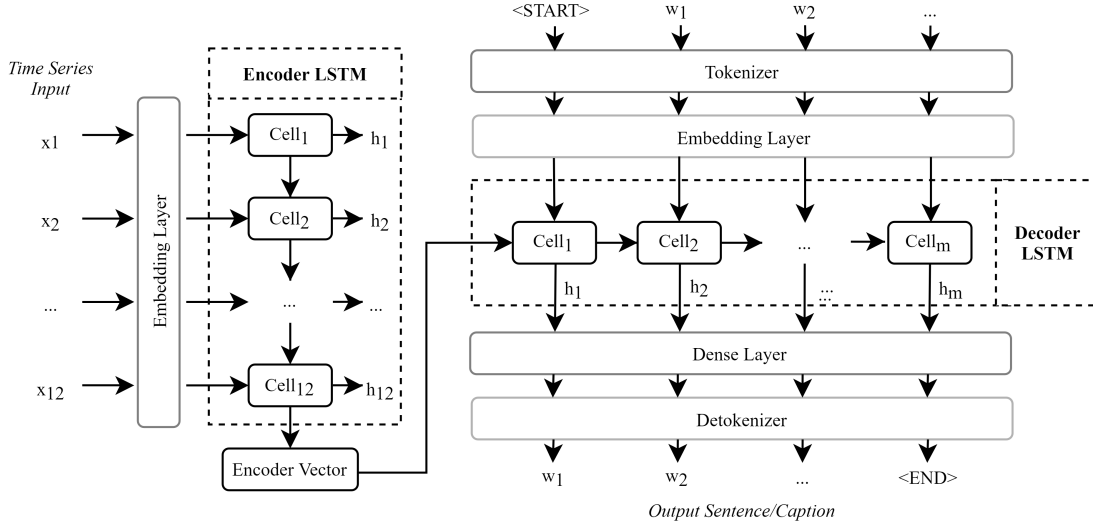


Figure 3: An illustration of our Encoder-Decoder LSTM system architecture.

The data collection process resulted in 306 captions, corresponding to three captions for each chart. The average length of a caption is 53.26 words. Splitting each caption into sentences, the dataset contains 1075 sentences with an average length of 14.84 words. An example of the collected captions is shown in Figure 2, within the annotation interface.

Data pre-processing: Once the data collection phase was completed, the captions needed some pre-processing to be suitable for effective learning. First, some data filtering and data cleaning procedures were applied to the collected captions, converting text to lower case, removing special characters and replacing punctuation with tokens (e.g. from “,” to “COMMA”). Next, the captions were generalized, in order to support the training of a system able to generate captions on possibly novel thematic areas and not restricted to the vocabulary acquired during annotation. In essence, we applied an edit distance function to calculate a Jaccard Similarity Index [7] between different N-Grams of each captions and specific terms like the title, the year and the unit of measurements associated with the data chart. Next, each sub-sequence within the captions that was sufficiently similar to a specific term was substituted with the corresponding pre-defined tokens. For example:

The *hardwood production* for *Canada* in *2018* declined in July.

The *TKN_About* for *TKN_Geo* in *TKN_Year* declined in July.

Note that captions also contain references to specific values of the time series they describe. These needed to be generalized as well. Our solution was to normalize all the values so that differences due to the time series evolving over different time ranges were eliminated.

3 MODELS, INPUTS AND OUTPUTS

In this study, besides the (Encoder-Decoder LSTM) neural model, we consider two simpler baselines (Bi-Gram and Similarity), which accomplish the task through non-neural, but still data-driven, approaches. Possible inputs for these models comprise a training set of normalized time-series with their associated captions, as well

as the normalized time-series data for which a caption should be generated. Each normalized time series is composed by 12 values x_i in the range $[1, 100]$, where each value is associated with a specific month of the time series, so for example x_1 with January, x_2 with February and so on. For each model type, we developed two variants: one generating one sentence about the chart, the other producing a sequence of sentences, i.e., a complete caption.

Model 1 - Bi-Gram Model: A statistical language model assigns a probability $P(w_1, \dots, w_m)$ to any sequence of m words. N-gram models [6] are basic language models, making the assumption that the probability of a word only depends on the previous n words. Once they are acquired, by simple counting in a dataset, they can also be used for generation [12]. In the Bi-Gram case, that we use in this baseline, the first word w_1 is sampled from the conditional probability $P(w_1 | <S>)$ (where $<S>$ is the start of sentence symbol), then the next word is sampled from $P(w_2 | w_1)$ and so on. This process continues until the end of sentence character is generated for sentence generation and until the end of caption character is generated for caption generation. Notice that this naive baseline does not consider the time series values at all, but exclusively bases its output on the vocabulary’s words conditional probabilities learned from the training set.

Model 2 - Similarity Model: In contrast to the naive Bi-Gram model, the Similarity model does consider the time series values in input. In fact, inspired by the nearest neighbor algorithm, it tries to match the input normalized time series ts_{input} with the most similar one within the training set ts_{sim} . Similarity is calculated using the Pearson coefficient, which measures linear correlation in the range $+1$ (total positive correlation) and -1 (total negative correlation), with 0 indicating no correlation.

Once the most similar time series ts_{sim} has been identified, the associated caption(s) are used for generation. More specifically, one of these captions is randomly selected and the generic predefined tokens that it contains are filled with the corresponding information from ts_{input} . A similar process is applied to generate a single

sentence, by randomly selecting a sentence from the captions associated with ts_{sim} and filling its generic tokens appropriately with information from ts_{input} .

Model 3 - Encoder-Decoder LSTM Model: This is a standard neural model composed by two LSTMs: an encoder and a decoder. The core concept of this model is the cell state, which can carry relevant information throughout the processing of the sequence; so even information from the earlier time steps can influence later time steps, reducing the effects of short-term memory [11]. One of the main advantages of this kind of model is that the input and the output sequences can be of different length.

As shown in Figure 3, the encoder LSTM accepts as input the sequence of values x_t of the time series and produces as output a vector of features called *Encoder Vector* which is fed into the decoder. The LSTM decoder predicts an output h_t at time step t which, by applying the function $softmax(W^S h_{t-1})$ and a detokenization process, allow to obtain the output word w_t .

Finally, in order to let this model be able to generate either a single sentence or a whole caption, we introduced specific tokens to indicate the start and the end of either a sentence or a caption.

3.1 Training Configuration

The final model was trained using the Adam optimizer with a learning rate of 0.001 and a drop out rate of 0.2. Other hyper-parameters were set to rather standard default settings. The batch size was set to 32, the embedding size to 128 and the size of the LSTM hidden states is 512. The model was trained for 300 epochs.

4 EVALUATION AND RESULTS

We run an experiment to compare the Encoder-Decoder Model (*M0*) against the naive Bi-Gram Model (*BS0*) and the Similarity Model (*BS1*), evaluating them on the two tasks of generating a single sentence and generating a complete caption. The models evaluation was conducted using a 5-Fold Cross-Validation [21] procedure, training each model on 80 time series (~240 captions) and testing on the remaining 20 (~60 captions). We used the ROUGE metrics [14], since they allow to compare sentences automatically produced by a system against a reference or a set of reference (human-produced) sentences. The results are summarized in Table 1 and 2. Due to space limitations, we report precision (*Pr*), recall (*Re*) and F1-Score (*F1*) on only some of the most informative ROUGE metrics; namely, *ROUGE-N* (1 and 2), which measure the n-grams overlap between the system and reference summaries, along with *ROUGE-L*, which identifies the Longest Common Sub-sequence, so capturing sentence structural similarity.

Although these results are quite preliminary, the neural model shows a clear advantage over the other models, dominating in both Sentence (see Table 1) and Caption Generation (see Table 2) on the key metric of Average F1 (which combines Precision and Recall). When we look in detail at Precision/Recall results, BS1 provides the best Precision in Caption Generation. However this does not lead to the highest F1. Furthermore, the neural model is the most stable model, showing a lower standard deviation in all the metrics, especially when compared with BS1. For lack of space, we can only show one sample of our model's output (see Figure 1). More examples generated by the three models will be added as auxiliary material at publication.

Model	Metric	AVG			Best		
		Pr %	Rec %	F1 %	Pr %	Rec %	F1 %
BS0	ROUGE-1	16.47	21.46	18.62	25.97	41.49	31.93
	ROUGE-2	2.87	4.18	3.40	11.30	17.91	13.83
	ROUGE-L	18.70	32.02	20.62	27.89	41.78	33.44
BS1	ROUGE-1	26.76	26.10	26.32	56.02	59.23	57.28
	ROUGE-2	8.90	8.76	8.79	33.06	35.52	34.07
	ROUGE-L	29.36	28.88	28.45	56.52	59.55	57.73
M0	ROUGE-1	30.36	31.19	30.75	51.54	58.72	54.84
	ROUGE-2	11.51	11.80	11.64	40.75	45.81	43.10
	ROUGE-L	33.09	33.81	33.44	61.80	70.14	65.67

Table 1: Automatic Evaluation results of Sentence Generation with a 5-Fold Cross Val using ROUGE (Average and Best). Comparing three different models BS0, BS1 and M0 considering Precision (*Pr*), Recall (*Rec*) and F1-Score (*F1*).

Model	Metric	AVG			Best		
		Pr %	Rec %	F1 %	Pr %	Rec %	F1 %
BS0	ROUGE-1	35.80	28.73	31.85	34.37	36.12	35.19
	ROUGE-2	9.56	8.37	8.91	13.14	12.68	12.88
	ROUGE-L	29.52	23.92	26.40	28.81	31.24	29.92
BS1	ROUGE-1	37.34	35.88	36.61	37.94	45.50	41.23
	ROUGE-2	12.27	12.10	12.15	15.76	18.30	16.91
	ROUGE-L	32.96	31.49	32.19	33.90	39.74	36.48
M0	ROUGE-1	28.00	43.55	39.57	32.87	54.81	41.08
	ROUGE-2	12.09	15.25	13.48	15.90	22.15	18.50
	ROUGE-L	31.56	38.02	34.48	30.63	47.36	37.16

Table 2: Automatic Evaluation results of Caption Generation with a 5-Fold Cross Val using ROUGE (Average and Best).

5 CONCLUSION

In this paper, we have studied data-driven approaches to the task of generating captions for information visualizations. We collected and used a seed version dataset containing time-series and corresponding line charts, which may be used as pilot for future works. The described experimental results demonstrate that our neural approach to the problem is the top performer (AVG. F1) for both the sentence and the caption generation tasks.

In the future, we plan to perform a detailed error analysis to better understand the strength and weaknesses of the different models. After that, we would like to focus on collecting a much larger corpus of visualization-caption pairs for training and testing, as well as exploring training the model with both features extracted from the time series data and from the bitmap image. Also, in addition to use automatic metrics, we plan to run an evaluation with humans, where participants would rank the outputs of different models. Next, to move beyond simple time-series and corresponding line charts, we plan to consider more sophisticated input representations that specify both the structure of the visualization and the data to be visualized, like the DRACO framework recently proposed in [17]. This will also likely require to apply more complex deep learning architectures, such as HAN [25], which combines hierarchical LSTM with attention mechanisms.

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